Constructive and Destructive Training Regimes: A Framework for Interpreting SGD Stability

Thiago Mattar, Cristiano L. Castro, Frederico Coelho, Yan G. Ferreira, Antonio P. Braga

***Abstract*—Parameter updates can reinforce or cancel one another even when training loss continues to decrease. An exact finite-time decomposition of stochastic gradient descent (SGD) displacement separates gradient magnitudes from interactions across all temporal lags. Consecutive-gradient cosine provides a local directional diagnostic, while normalized displacement measures cumulative retention of traveled distance. Local expansions connect these quantities to curvature and sampling noise without identifying alignment sign with stability or objective progress. Full-batch experiments exhibit gradient reversal alongside continued loss reduction. Matched mini-batch continuations show that sampling changes both gradient estimates and the paths traversed through the objective. Under Adam, raw-gradient diagnostics remain informative about the loss landscape traversed by adaptive steps. Controlled full-batch experiments reveal loss spikes associated with pre-rise gradient reversal and norm amplification. Loss-defined events quantify their magnitude and recovery, while descriptive regressions relate the initial regime and subsequent norm growth to relative peak loss. The resulting framework supports the formulation and evaluation of informed learning-rate interventions; it does not prescribe an alignment-maximizing schedule.**

***Index Terms*—Stochastic Gradient Descent, Adam, Gradient Alignment, Parameter Displacement, Learning-Rate Scheduling**

I. Introduction

Neural scaling laws have ushered in new paradigms for interpreting deep learning dynamics. By establishing that performance gains follow predictable behavior relative to model size, dataset scale, and compute, stochastic optimization methods have been pushed toward increasingly extended training horizons while operating under tight efficiency constraints [1]–[4]. In this regime, optimization stability becomes an important practical requirement, since unstable training may result in slow convergence and high sensitivity to hyperparameter choices. In practice, stability is often maintained through a set of heuristics, such as momentum, batch-size scaling, normalization, and learning-rate schedules [5]–[9]. Yet, despite their effectiveness, it is not fully understood why these mechanisms work, and under what conditions they fail.

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Classical local analyses relate the permissible step size to curvature in sharp directions of the loss [10], [11]. Stochastic extensions, including the break-even point framework, also account for curvature variation induced by sampling [12]. The Edge-of-Stability literature shows that deep networks can continue to make progress while oscillating near, or beyond, the nominal sharpness threshold [13]–[15]. These results motivate a complementary diagnostic question: how do the magnitudes and directions of successive gradients combine into parameter motion along the observed training trajectory? Gradient information already computed during optimization makes this question accessible without estimating the full Hessian spectrum or gradient-noise covariance at every update.

To examine these interactions, we describe parameter motion through gradient magnitudes and directional alignment. Starting from an untrained reference, training changes the parameters through which the model expresses what it has learned. Displacement provides a geometric description of that change; it does not quantify retained knowledge or establish whether the change improves the objective. Positive and negative gradient interactions respectively reinforce and cancel components of the accumulated motion. Their effect must be evaluated at the relevant temporal horizon and distinguished from loss reduction.

We develop an exact finite-time decomposition of SGD displacement that retains all cross-step interactions. Consecutive-gradient cosine supplies a local online diagnostic, while normalized displacement measures the fraction of accumulated step length retained as net motion. Local expansions relate gradient changes to directional curvature and sampling noise. The distinction between these quantities is consequential: a trajectory can reverse successive gradient directions while continuing to reduce loss, and sampling can change both the measured alignment and the subsequent path.

The same distinction becomes essential under adaptive optimization. Adam already computes raw gradients and moment estimates [16]; its parameter updates combine temporal averaging, coordinate scaling and the learning rate. Monitoring these successive quantities can reveal changes that a single alignment statistic would obscure. Raw-gradient diagnostics remain informative about the traversed loss landscape even though Adam transforms those gradients before updating the parameters. Controlled full-batch trajectories reveal repeated loss excursions associated with gradient reversal and norm amplification. We examine individual spikes, their population anatomy and the association between the pre-rise regime, subsequent norm growth and peak magnitude. These observations motivate interventions while distinguishing descriptive

evidence from a causal mechanism or a validated prediction rule.

Our contributions are:

- A displacement-based account of local and cumulative gradient interaction, with explicit connections to curvature, sampling noise and finite-step loss changes;
- Controlled full-batch and mini-batch experiments that distinguish directional reversal, displacement retention and objective progress;
- A practical extension to Adam that separates raw-gradient information, momentum and adaptive directions, supported by controlled training and an exploratory analysis of loss-spike anatomy across models and rates.

The remainder of the paper is organized as follows. Section II reviews training-stability perspectives. Section III develops the displacement formulation and its diagnostics. Section IV examines full-batch GD and mini-batch sampling. Section V extends the interpretation to Adam and outlines the empirical requirements for informed scheduling. Section VI concludes.

II. Background: Stability Analysis of SGD

Training stability concerns how perturbations and loss evolve under repeated updates. Overshoot and oscillation are useful symptoms, but can also accompany sustained objective progress. Understanding these dynamics requires examining both the *local curvature* of the loss landscape and the *stochasticity* introduced by mini-batch sampling [10], [11], [13].

A. Local Curvature of the Loss Function and Stability

Training can be interpreted as a discrete-time dynamical system in which the local curvature of the loss function determines the stability of the updates [10], [11]. The curvature indicates how sensitive the loss is to parameter changes in different directions: when curvature is small, updates tend to produce smooth descent toward a minimum, whereas in directions of high curvature the same step can overshoot the minimum and induce oscillatory behavior [10], [11], [17]. For this reason, classical analyses of gradient descent relate stability to the spectrum of the Hessian of the loss function, whose eigenvalues quantify directional curvature and indicate how sensitive optimization is to the choice of learning rate [10], [11].

Under this view, let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t, and let g_t denote a mini-batch gradient evaluated at θ_{t-1}. For a constant learning rate $\eta > 0$, the SGD update is

$$\theta_t = \theta_{t-1} - \eta\, g_t. \tag{1}$$

For full-batch GD, $g_t = \nabla\mathcal{L}(\theta_{t-1})$, where $\mathcal{L}$ denotes the full training objective. Defining $H_t := \nabla^2\mathcal{L}(\theta_{t-1})$, the Jacobian of the deterministic update map is

$$J_t = I - \eta\, H_t. \tag{2}$$

Consider a small perturbation e_{t-1} around θ_{t-1}. Under the local linearization of the full-batch GD update, this perturbation evolves as

$$e_t \approx J_t e_{t-1} = (I - \eta H_t) e_{t-1},$$

In a locally quadratic regime near a minimum, the Hessian is approximately constant and admits eigenpairs (λ_i, v_i). Decomposing the perturbation as $e_t = \sum_i \delta_t^{(i)} v_i$, each eigendirection evolves independently according to

$$\delta_t^{(i)} = (1 - \eta\lambda_i)\, \delta_{t-1}^{(i)}.$$

Thus, the factor $1 - \eta\lambda_i$ determines whether perturbations along direction v_i decay or grow. For a positive-definite Hessian, a classical necessary and sufficient condition for linear stability is

$$\max_i |1 - \eta\lambda_i| < 1 \;\Leftrightarrow\; \eta\,\lambda_{\max}(H_t) < 2, \tag{3}$$

which means that all perturbation components contract over time [10], [11]. In other words, if $0 < \eta\lambda_i < 1$, the perturbation decays monotonically along v_i; if $1 < \eta\lambda_i < 2$, it alternates sign while shrinking, producing oscillations around the minimum; and if $\eta\lambda_i > 2$, it alternates sign while growing, leading to divergence. In over-parameterized networks, however, the Hessian spectrum is typically highly anisotropic, with a small number of outlier directions carrying most of the sharpness, while the bulk remains near zero [18], [19]. Although this suggests monitoring $\lambda_{\max}(H_t)$ as a stability diagnostic, estimating leading Hessian eigenvalues online in large neural networks is computationally expensive, and mini-batch estimates provide only noisy, pointwise proxies of the local geometry.

B. Mini-Batch Stochasticity

A complementary line of work studies SGD through the lens of stochastic optimization, interpreting the mini-batch gradient as a noisy approximation of the full gradient computed over the entire dataset [20]–[23]. In this sense, each SGD step combines a deterministic descent component with a zero-mean random fluctuation induced by sampling, which covariance provides information about how noise is distributed across parameter-space directions and how it may affect trajectory stability. Formally, under fresh, conditionally unbiased sampling, we write the mini-batch gradient as

$$\begin{aligned} g_t &= \nabla\mathcal{L}(\theta_{t-1}) + \xi_t, \\ \mathbb{E}[\xi_t \mid \theta_{t-1}] &= 0, \\ \mathrm{Cov}(\xi_t \mid \theta_{t-1}) &\approx \frac{1}{b}\,\Sigma(\theta_{t-1}), \end{aligned} \tag{4}$$

where ξ_t is the sampling noise, b is the batch size, and $\Sigma(\theta_{t-1})$ denotes the covariance matrix of per-sample gradients around the full gradient. At the current parameters, write $h_i := \nabla\ell_i(\theta_{t-1})$, where ℓ_i is the loss for sample i, and $\mu := \frac{1}{N}\sum_{i=1}^N h_i = \nabla\mathcal{L}(\theta_{t-1})$, where N is the dataset size. The covariance matrix is

$$\Sigma(\theta_{t-1}) := \frac{1}{N}\sum_{i=1}^{N}\big(h_i - \mu\big)\big(h_i - \mu\big)^\top, \tag{5}$$

and the scalar dispersion of per-sample gradients around the full gradient is given by the trace of $\Sigma(\theta_{t-1})$

$$\mathrm{tr}(\Sigma(\theta_{t-1})) = \frac{1}{N}\sum_{i=1}^{N}\|h_i - \mu\|^2.$$

This makes clear why $\Sigma(\theta_{t-1})$ is generally not a viable online diagnostic: computing it requires access to the full gradient μ and all per-sample gradients $\{h_i\}_{i=1}^N$ at each training step, which is impractical and incurs at least a full-pass cost per step.

C. Curvature–Noise Interaction

The *break-even point* framework of Jastrzebski et al. [12] is an attempt to extend the classical curvature-based stability condition to SGD. The central idea is to analyze stability along the sharpest direction of the loss landscape while explicitly accounting for mini-batch stochasticity. More precisely, the framework proposes a one-dimensional necessary-and-sufficient condition for SGD stability at iteration t,

$$(1 - \eta\lambda_{\max}(H_t))^2 + s_t^2 \frac{\eta^2(N-b)}{b(N-1)} \;\leq\; 1, \tag{6}$$

where $\lambda_{\max}(H_t)$ is the top Hessian eigenvalue, N is the dataset size, b is the batch size, and s_t^2 denotes the variance across training examples of the curvature measured along the leading Hessian direction. Large s_t^2 means different mini-batches see substantially different local curvature, while small s_t^2 means they see similar curvature. In this sense, Eq. (6) can be read as a stochastic extension of the deterministic stability condition in Eq. (3), so that the first term captures curvature-induced amplification, while the second captures the destabilizing effect of mini-batch stochasticity.

Under this definition, the break-even point is defined as the first iteration at which the left-hand side of Eq. (6) reaches 1. Intuitively, it marks the end of an early phase during which the choice of learning rate and batch size can strongly shape the later optimization trajectory. In this sense, a central conjecture of this framework is that, prior to this point, hyperparameters largely determine the maximum curvature and stochasticity levels encountered later in training, thereby setting the regime in which the subsequent dynamics unfold.

Although the local analysis simplifies both the interpretation and the mathematical treatment of the stability problem, it is subject to well-known limitations. For instance, the Edge-of-Stability (EoS) phenomenon challenges the local-quadratic picture even in deterministic gradient descent (GD). Across architectures and tasks, full-batch GD often exhibits progressive sharpening, with the largest eigenvalue of the Hessian, $\lambda_{\max}(H_t)$, rising until it hovers near (and sometimes above) the nominal threshold $2/\eta$ (Eq. 3), while optimization continues to make progress despite short-term non-monotonicity [13]–[15]. Thus, successful training can coexist with local linear instability for extended periods, indicating that sharpness alone does not fully explain and determine trajectory-level behavior.

Together, these results clarify both the value and the limitation of curvature–noise interaction criteria. Quantities such as the leading Hessian eigenvalue $\lambda_{\max}(H_t)$ and the gradient-noise covariance $\Sigma(\theta_{t-1})$ characterize local geometry and stochasticity, but can be expensive to estimate online. Their pointwise values also leave open how successive updates interact and accumulate over a training trajectory. This motivates complementary diagnostics that describe directional reinforcement and cancellation using quantities available during optimization.

D. Practical Stability Analysis in Large-Model Pretraining

Large-model training reports approach stability through observable symptoms and interventions on the training recipe. An instructive example is PaLM: Section 5.1 reports approximately 20 loss spikes during training of the 540B model despite global gradient clipping. The authors recovered by returning to a checkpoint about 100 steps before a spike and skipping 200–500 batches. Replaying those batches from a different parameter state did not reproduce the spikes, suggesting an interaction between data and model state rather than a property of the batches alone [24].

Reports accompanying released model weights provide further operational evidence. During OPT-175B training, loss divergence was associated with failures of the dynamic loss scale and growing final-layer activation norms. Recovery involved reducing the learning rate and restarting from an earlier checkpoint; lowering the gradient-clipping threshold from 1.0 to 0.3 also improved stability early in training [25]. These observations motivate monitoring the numerical state of training alongside the loss.

OLMo 2 makes this diagnostic process explicit by measuring a rolling spike score on training loss and gradient norm, and using controlled ablations to study initialization, normalization, and optimizer settings. Its revised recipe combines global gradient clipping with changes including query–key normalization, reordered normalization, and a logit regularizer (z-loss) [26]. In Llama 3, the authors report a stable pretraining recipe with few loss spikes and no interventions for divergence; they use learning-rate warmup and a staged batch-size schedule, and emphasize FP32 gradient accumulation and reduction for numerical stability [27]. These reports document both failure recovery and recipes that avoid recurrent failures, rather than a universal instability mechanism.

Loss, gradient norm, and activation statistics are practical monitoring signals, but gradient magnitude alone does not describe the relationship between successive directions. A large gradient can reinforce or oppose its predecessor, and a sequence of individually bounded steps can still exhibit cumulative cancellation. We therefore seek a complementary description that can be computed from training gradients and interpreted through their temporal geometry. The following section develops parameter displacement and consecutive-gradient cosine similarity for this purpose, connecting the latter to directional curvature and gradient noise without requiring online Hessian eigendecomposition.

III. Characterizing Constructive and Destructive Regimes

Loss spikes and oscillatory updates motivate practical diagnostics that can be computed from gradients already available during training. We develop these diagnostics by relating parameter displacement, gradient magnitudes and alignment,

curvature and sampling noise. The proposed quantities characterize constructive and destructive regimes, and inform possible interventions without the additional overhead of computing Hessians.

Consider the plain SGD update in Eq. (1), with constant learning rate $\eta > 0$. As before, $\theta_t \in \mathbb{R}^p$ denotes the parameters after update t, and g_t is the mean mini-batch gradient evaluated at θ_{t-1}. Summing the updates gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{7}$$

For a model trained from scratch, θ_0 represents the initialized model that has not yet learned from the training data. After T updates, the trained parameters θ_T encode what the model has learned within that training budget. The displacement Δ_T captures the net parameter change produced by training. Expanding the squared displacement separates gradient magnitudes from the interactions between different steps

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t. \end{aligned} \tag{8}$$

The terms $\|g_t\|^2$ are nonnegative and determine the squared lengths of individual SGD steps through $\eta^2\|g_t\|^2$. In full-batch GD, a larger gradient norm indicates a steeper local loss and, therefore, a larger first-order decrease along the negative gradient. The cross-step terms $g_u^\top g_t$ describe how these steps interact – positive inner products reinforce the displacement, whereas negative inner products cancel part of it. Large gradient norms can therefore coexist with cancellation along the training trajectory, and destructive interference (i.e. persistent $g_u^\top g_t < 0$) can limit displacement growth.

To isolate directional alignment from gradient magnitudes, define the cosine similarity between any two nonzero gradients and its consecutive-step form

$$\rho_{u,t} := \frac{g_u^\top g_t}{\|g_u\|\|g_t\|}, \quad 1 \le u < t \le T. \tag{9}$$

In this setup, the alignment between consecutive gradients is $\rho_t := \rho_{t-1,t}$, $t = 2, \ldots, T$. Substituting into Eq. (8) expresses the squared displacement entirely through gradient norms and directional alignment

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} \|g_u\|\|g_t\|\rho_{u,t}. \end{aligned} \tag{10}$$

For fixed gradient norms, positive $\rho_{u,t}$ increases the squared net displacement, negative $\rho_{u,t}$ contributes cancellation, and orthogonal gradients have no pairwise contribution. In practice, ρ_t provides an online diagnostic of the interaction between successive updates: persistent positive or negative values characterize constructive or destructive regimes.

A. Directional Curvature and Consecutive Alignment

For mini-batch SGD, we explicitly incorporate stochastic gradient noise in Eq. (4), $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$. Suppose the full objective $\mathcal{L}$ is twice continuously differentiable along the preceding step, and write $H_{t-1} := \nabla^2\mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} and substituting the preceding SGD update yields

$$g_t = (I - \eta H_{t-1})g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{11}$$

where r_t is the Taylor remainder. Neglecting r_t recovers the first-order approximation. Taking the inner product with g_{t-1} then gives

$$\begin{aligned} \|g_{t-1}\|\|g_t\|\rho_t \approx \underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ \underbrace{- \eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} \\ \underbrace{+ g_{t-1}^\top(\xi_t - \xi_{t-1})}_{\text{Noise contribution}}. \end{aligned} \tag{12}$$

Positive curvature reduces the component of the next gradient along the preceding direction. Its strength relative to the preceding squared norm is the directional curvature

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \tag{13}$$

Thus, κ_t measures how strongly a step changes the gradient along the direction actually followed, whereas ρ_t measures the resulting alignment. Sampling modifies this response through the normalized noise interaction

$$\varepsilon_t := \frac{g_{t-1}^\top(\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{14}$$

Dividing Eq. (12) by the two gradient norms gives

$$\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t). \tag{15}$$

In full-batch GD, $\varepsilon_t = 0$. Within this first-order approximation, $\eta\kappa_t < 1$ retains a positive projection, while $\eta\kappa_t > 1$ reverses it: the next update partially opposes the preceding step. At $\eta\kappa_t = 1$, the new gradient has lost its component along the old direction. It is orthogonal if nonzero; if it vanishes, cosine is undefined. The norm ratio determines how this projected change translates into the magnitude of ρ_t.

The threshold at one also has an optimization interpretation. Along the full-gradient direction, the local quadratic loss model is $\mathcal{L}(\theta_{t-2} - \alpha g_{t-1}) \approx \mathcal{L}(\theta_{t-2}) - \alpha\|g_{t-1}\|^2 + \alpha^2\kappa_t\|g_{t-1}\|^2/2$. For $\kappa_t > 0$, its directional minimum is at $\alpha = 1/\kappa_t$. Negative alignment therefore indicates that the step has passed this directional minimum under the approximation. Other gradient components may remain: this threshold identifies directional overshoot, rather than a stationary point or the loss-increase boundary derived below.

This interpretation leads to a measurable diagnostic. Rearranging Eq. (15) gives

$$\eta\widehat{\kappa}_t := 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \approx \eta\kappa_t - \varepsilon_t. \tag{16}$$

The left-hand side uses only consecutive gradients: it measures their finite-step projected change without a Hessian evaluation. In full GD it estimates directional curvature when the Taylor remainder is small; in SGD it also contains sampling noise. It characterizes the realized step, whereas Eq. (3) concerns perturbations along the most curved local direction.

B. Noise Contribution under Mini-Batch Sampling

Unbiased sampling does not make the noise interaction vanish. Write $g_{t-1} = \nabla\mathcal{L}(\theta_{t-2}) + \xi_{t-1}$. Fresh, conditionally unbiased batches give $\mathbb{E}[g_{t-1}^\top \xi_t] = 0$ and $\mathbb{E}[\nabla\mathcal{L}(\theta_{t-2})^\top \xi_{t-1}] = 0$. With finite second moments, the preceding gradient contributes its own noise energy:

$$\mathbb{E}[g_{t-1}^\top(\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{17}$$

Thus, even unbiased sampling noise contributes a systematic nonpositive term to the expected unnormalized interaction between consecutive gradients.

This contribution is directly related to mini-batch size. Consider a mini-batch of b examples sampled uniformly without replacement from $N > 1$ training examples. Using the per-sample gradient covariance $\Sigma(\theta)$ defined in Eq. (5), the sampling variance satisfies

$$\mathbb{E}[\|\xi_{t-1}\|^2] = \frac{N-b}{b(N-1)}\mathbb{E}[\operatorname{tr}\Sigma(\theta_{t-2})]. \tag{18}$$

At fixed parameters, smaller batches produce larger sampling variance and, consequently, a larger negative expected noise contribution. For full-batch gradients ($b = N$), this contribution vanishes.

Noise can weaken directional coherence by enlarging gradient norms relative to their shared signal. For independent estimates at two fixed states, let μ, ν be the nonzero full gradients and v_μ, v_ν the expected squared noise norms. When sampled norms concentrate,

$$\mathbb{E}[\cos(\mu+\xi, \nu+\zeta)] \approx \frac{\cos(\mu,\nu)}{\sqrt{(1+v_\mu/\|\mu\|^2)(1+v_\nu/\|\nu\|^2)}}.$$

This attenuation moves either sign toward zero. At the same fixed state, the common full-gradient direction gives the special case

$$\mathbb{E}[\rho_t] \approx \frac{\|\nabla\mathcal{L}(\theta)\|^2}{\|\nabla\mathcal{L}(\theta)\|^2 + \frac{N-b}{b(N-1)}\operatorname{tr}\Sigma(\theta)}. \tag{19}$$

Here the expected cosine is nonnegative; the approximation quantifies the dilution of a common direction by sampling. During SGD, a sampled update also changes the next state, coupling noise and curvature. The fixed-state attenuation argument therefore isolates a mechanism, while the batch-size intervention in Section IV-D tests its expression along evolving trajectories.

C. Normalized Displacement and Cumulative Interference

While the consecutive cosine similarity ρ_t measures local lag-1 directional alignment, it does not capture interactions between more distant updates (lag$\ge$ 2). To measure their cumulative effect, we define the normalized displacement

$$R_T := \frac{\|\Delta_T\|}{\eta\sum_{t=1}^T \|g_t\|} = \frac{\|\sum_{t=1}^T g_t\|}{\sum_{t=1}^T \|g_t\|}, \qquad 0 \le R_T \le 1, \tag{20}$$

For constant-step plain SGD, the denominator represents the total distance traveled in parameter space, while the numerator measures the net displacement from initialization. Thus, R_T quantifies the fraction of accumulated step length retained as net displacement.

Using Eq. (10), this ratio satisfies

$$R_T^2 = \frac{\sum_{t=1}^T \|g_t\|^2 + 2\sum_{1\le u<t\le T}\|g_u\|\|g_t\|\rho_{u,t}}{\left(\sum_{t=1}^T \|g_t\|\right)^2}. \tag{21}$$

Unlike ρ_t, which considers only adjacent gradients, R_T incorporates interactions across all temporal lags. When R_T is near one, almost all traveled distance survives as net displacement; when it is near zero, little does. Repeated back-and-forth steps and weakly coherent sampled directions both suggest low retention. For example, equal-length orthogonal updates give $R_T = 1/\sqrt{T}$, whereas equal opposing pairs cancel exactly. We therefore expect small R_T when such behavior persists over a substantial part of an SGD path. Its value is relative to traveled distance: absolute displacement can still grow, and the sign of the complete interaction in Eq. (21) requires the full decomposition.

D. Directional Overshoot and Loss Increase

Directional overshoot need not undo the loss decrease already made along a step. Define its full-objective change as $\delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. The second-order approximation derived in Appendix A separates three contributions:

$$\begin{aligned}\delta\mathcal{L}_{t-1} \approx\ & \underbrace{-\frac{\eta}{2}\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{-\frac{\eta}{2}\rho_t\|g_{t-1}\|\|g_t\|}_{\text{Directional interference}} \\ & + \underbrace{\frac{\eta}{2}g_{t-1}^\top(\xi_{t-1}+\xi_t)}_{\text{Noise interaction}}.\end{aligned} \tag{22}$$

The magnitude term favors descent. The interference term reinforces it for positive alignment and opposes it for negative alignment: destructive interference contributes positively to the predicted loss change. With negligible noise and nonzero gradients, the approximation predicts a loss increase precisely when

$$\rho_t < -\frac{\|g_{t-1}\|}{\|g_t\|}. \tag{23}$$

This requires both negative alignment and gradient-norm growth. Neither alone suffices: the opposing component of the new gradient must exceed the preceding gradient magnitude. Large norms can therefore accompany descent, just as negative cosine can accompany a smaller loss decrease.

Substituting Eq. (15) with zero noise gives

$$\delta\mathcal{L}_{t-1} \approx \frac{\eta}{2}\|g_{t-1}\|^2(\eta\kappa_t - 2). \tag{24}$$

The two thresholds distinguish overshooting the directional minimum ($\eta\kappa_t > 1$) from overshooting far enough to raise the loss ($\eta\kappa_t > 2$). Between them, the gradients oppose each other while the step still improves the objective. These are local quadratic predictions; finite-step nonlinear effects and the explicit SGD noise term can change the observed sign. A transient increase becomes a loss peak only if subsequent updates recover.

IV. Empirical Characterization of Gradient Interference

We use full-batch GD to characterize directional overshoot, its effect on cumulative motion, and its coexistence with loss reduction. We then vary batch size to examine how sampling changes these signatures.

The comparison covers a tanh MLP on 5,000 CIFAR-10 images, ResNet-32 on all 50,000 CIFAR-10 training images, and a word-level Transformer on WikiText-2. Each minimizes its complete fixed training objective for 3,000 updates at three constant learning rates (Table I). Within each model, rates share one frozen initialization and data preparation; they were selected by earlier training calibrations to expose contrasting dynamics, without test-loss selection. Plain GD uses no momentum, weight decay, clipping or schedule. These comparisons test whether the same geometric signatures recur across the three objectives; they do not isolate architecture effects. Appendix B specifies objectives, architectures, precision and measurement details.

A. A Gradient-Based View of Oscillatory Descent

Figure 1 pairs loss with the projected gradient change $\eta\widehat{\kappa}_t$. The higher-rate trajectories develop oscillatory loss while continuing to improve over longer timescales; their late projected change approaches two. Equation (24) explains why this value is informative: in the local quadratic model, it separates a loss-decreasing step from one that reverses the preceding progress. The observed pattern gives a gradient-based view of the oscillatory descent associated with the Edge of Stability [13].

The diagnostic requires consecutive gradients rather than Hessian–vector products or an eigensolver. It estimates curvature along the realized step when the local approximation applies; it does not estimate the largest Hessian eigenvalue that defines spectral EoS. This distinction permits a practical interpretation without treating the two quantities as interchangeable. The same local model explains the regime but does not reliably predict every nonlinear loss fluctuation: the stepwise approximation check and spectral coverage are retained in Appendix B.

B. Reversal, Step Magnitude and Displacement Retention

Figure 2 shows how the oscillatory regime changes parameter motion. Persistent negative cosine means that successive updates repeatedly oppose the preceding direction, consistent with overshooting its local directional minimum. The lowest MLP and Transformer rates retain positive alignment, whereas the other seven conditions approach nearly opposing gradients. In the MLP and ResNet, the transition also accompanies growing gradient norms: the steps become larger even as their directions increasingly cancel. These fluctuations arise without mini-batch sampling.

The decreasing R_t shows that larger steps need not translate into proportionally larger net motion. Repeated reversals consume path length, so the retained fraction can become small even while parameters continue to move away from initialization. This is a geometric inefficiency, not evidence of slower optimization. Table I makes the distinction concrete: within every model, the highest tested rate has the lowest retention, the largest absolute displacement, and the lowest final training loss. A rate can therefore be large relative to local directional minima yet productive over the full training horizon.

Adjacent cancellation also does not determine the sign of the complete cross-step interaction. That total is positive at the endpoint in all nine trajectories, including the seven with negative accumulated lag-one interaction (Table **??**). Longer-range reinforcement can coexist with repeated local reversal. Together, ρ_t, gradient norms and R_t distinguish local overshoot, step size and retained motion; loss establishes whether this motion improves the objective.

C. Oscillation Beyond Adjacent Updates

To examine the temporal extent of reversal, we fix one endpoint gradient and compare it with preceding gradients at every lag $k = 1, \ldots, 256$. Figure 3 uses the largest tested rate for each model and the common endpoint after 2,256 updates. Near-zero similarity at some lags does not imply that directional memory has disappeared: strongly opposing and reinforcing pairs can reappear at longer separations. The varying amplitude and sign pattern show that oscillation extends beyond a single adjacent pair, while its persistence depends on the model and part of the trajectory. This temporal structure explains why a lag-one statistic cannot summarize the complete interference term.

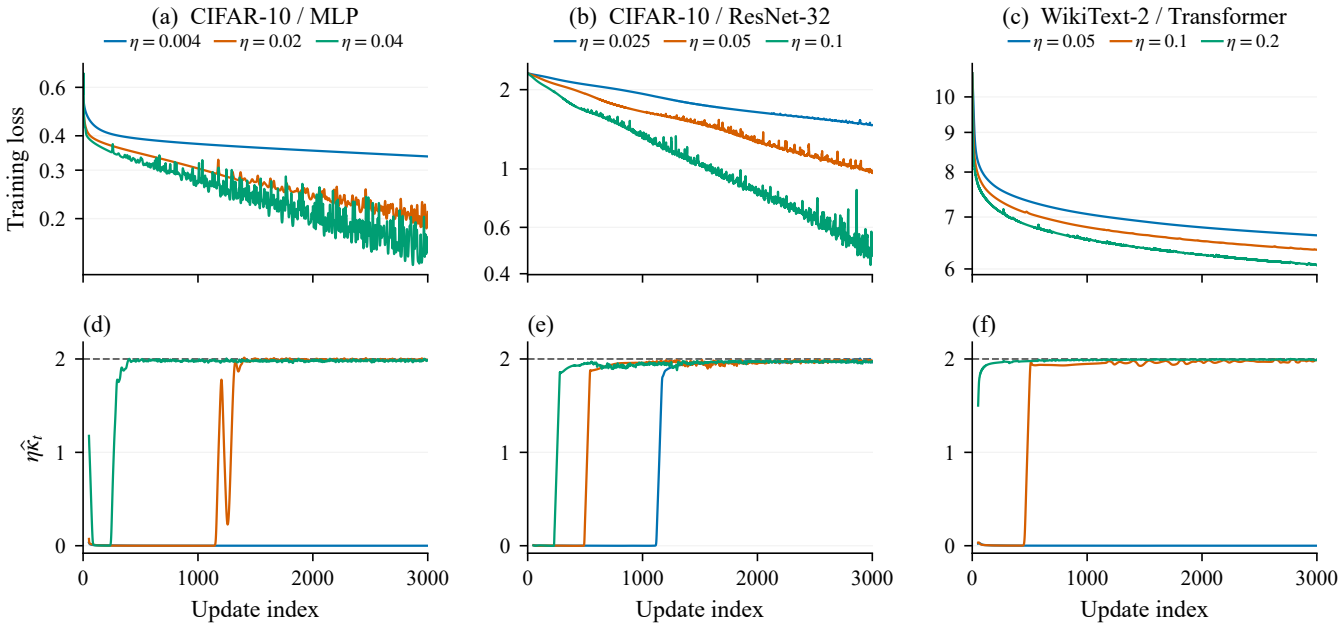


Fig. 1: Training loss and projected gradient change under full-batch GD. Columns identify the dataset/model; colors identify learning rates within each column. Top: unsmoothed full-objective loss on logarithmic axes. Bottom: complete 50-pair trailing means of the pointwise $\eta\widehat{\kappa}_t$, with the local quadratic loss-change boundary at two. Loss at index t is evaluated at θ_t; gradient diagnostics use g_t at θ_{t-1}. Each model uses one shared initialization. Smoothing summarizes the regime and does not identify its exact onset.

TABLE I: Endpoints at $T = 3000$ updates. The late means $\overline{\rho}_{\text{late}}$ and $\overline{\eta\widehat{\kappa}}_{\text{late}}$ average gradient indices $t = 2001, \ldots, 3000$. Displacement is $\|\Delta_T\| = \eta\|\sum_{t=1}^{T} g_t\|$, subject to parameter-update roundoff. Each model shares one initialization across rates; comparisons are within models. *Positive value below 0.00005, rounded to zero.

Dataset	Model	η	$\mathcal{L}(\theta_T)$	$\overline{\rho}_{\text{late}}$	R_T	$\overline{\eta\widehat{\kappa}}_{\text{late}}$	$\|\Delta_T\|$
CIFAR-10	MLP	0.004	0.3365	1.0000	0.7941	0.0000*	0.9511
		0.02	0.2114	-0.9913	0.0488	1.9924	3.1391
		0.04	0.1539	-0.9799	0.0257	1.9820	4.7628
	ResNet-32	0.025	1.4665	-0.9723	0.1385	1.9740	5.7045
		0.05	0.9672	-0.9708	0.0616	1.9746	9.0183
		0.1	0.4898	-0.9684	0.0374	1.9707	13.9329
WikiText-2	Transformer	0.05	6.6301	1.0000	0.7223	0.0002	11.3183
		0.1	6.3519	-0.9749	0.1395	1.9754	15.6026
		0.2	6.0697	-0.9937	0.0541	1.9937	21.9693

D. Mini-Batch Noise and Directional Coherence

We vary batch size at initialization and after 2,000 GD updates in the same three model–dataset cases, holding $\eta = 0.02$, 0.1 and 0.2 fixed for the MLP, ResNet-32 and Transformer, respectively. Each starting state receives 200-update continuations, with three initializations, eight sampling repetitions per stochastic condition and one full-GD control per initialization. Figure 4 compares mean sampled consecutive-gradient cosine over each continuation. Equal update budgets imply unequal data exposure; sampling units, objective coverage and numerical checks are in Appendix B.

The smallest batches weaken trained-state reversal in all three cases. Initial-state responses are less uniform: full GD has positive mean cosine in the MLP and ResNet and strongly negative mean cosine in the Transformer, where reversal develops within the first 200 updates. The MLP's sampled paths can also acquire negative mean alignment while its full-GD mean is positive. Sampling therefore changes the regime encountered along a path, in addition to perturbing gradients at fixed parameter states.

For the Transformer at $b = 8$, both starting states have mean cosine near -0.26, yet mean full-data loss falls from initialization and rises after GD training. Mean adjacent alignment alone therefore does not order training-objective progress.

The MLP's smallest trained-state batch likewise raises mean full-data loss, whereas full GD lowers it in both models.

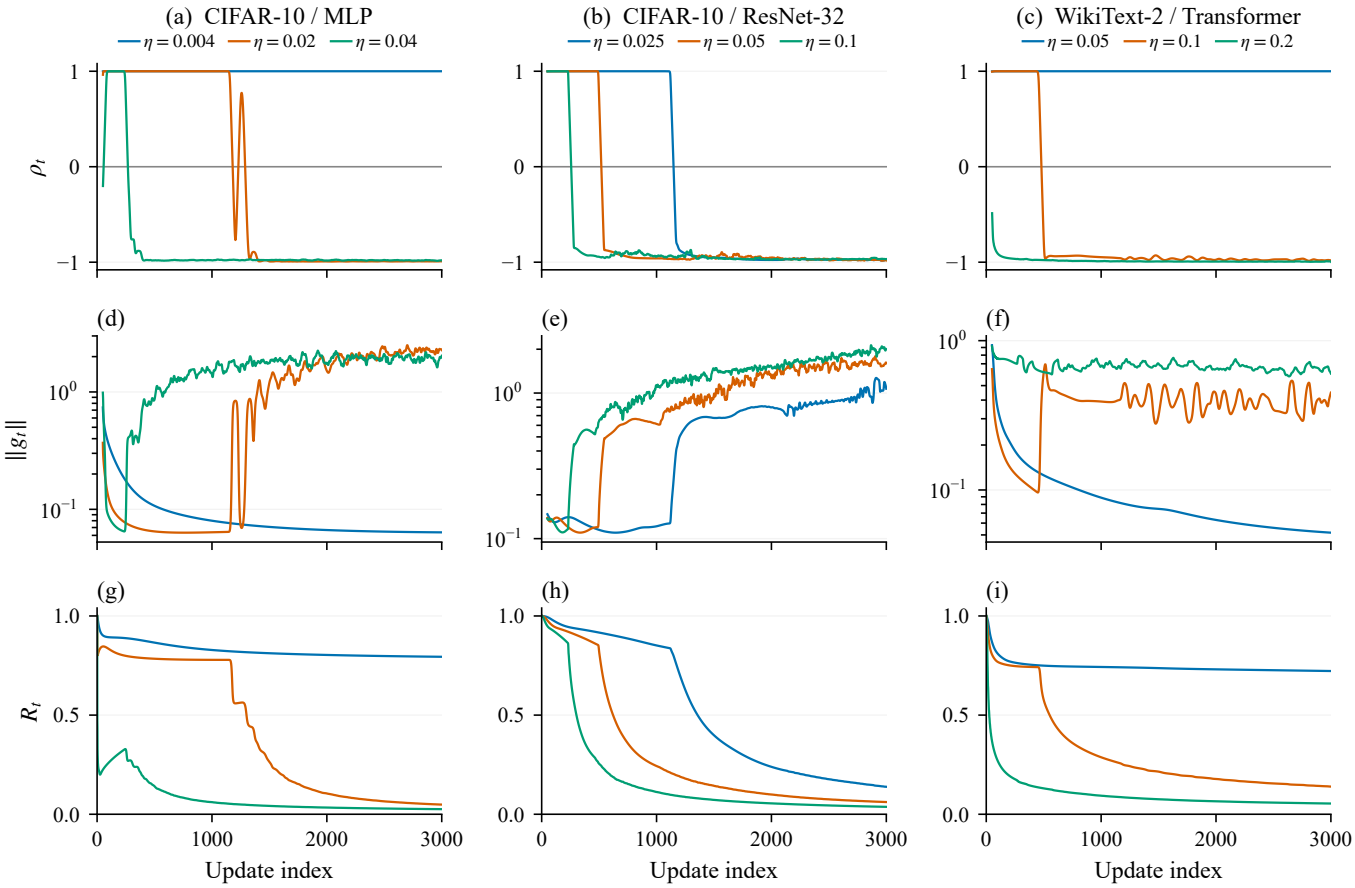


Fig. 2: Directional coherence and cumulative motion for the conditions in Fig. 1. Rows show consecutive cosine, gradient norm (logarithmic scale), and normalized displacement. Cosines and norms use complete 50-observation trailing means; R_t is the unsmoothed ratio over the complete prefix through update t. Colors and column ordering match Fig. 1. Raw cosine traces appear in Appendix B.

Noise covariance can affect the solutions favored by training [28], but mean cosine alone identifies neither that bias nor a generalization benefit. These comparisons describe selected objectives, rates and update budgets; they do not select an optimal batch size.

V. Raw-Gradient Diagnostics under Adam

Adam [16] transforms the raw gradient through first- and second-moment estimates,

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \end{aligned} \tag{25}$$

with zero initial moments. Bias correction gives $\widehat{m}_t = m_t/(1-\beta_1^t)$ and $\widehat{v}_t = v_t/(1-\beta_2^t)$, followed by

$$d_t = \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}, \qquad \theta_t = \theta_{t-1} - \eta d_t, \tag{26}$$

where operations are coordinatewise. Although d_t determines the step, the raw gradients still reveal variation in local slope along the adaptive trajectory. Their consecutive alignment and magnitudes distinguish directional reversal from a sharp increase in local slope. The first moment shows how this changing input propagates through the optimizer's memory.

The projected-change diagnostic remains exactly computable as

$$\eta \widehat{\kappa}_t = 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|}. \tag{27}$$

Under Adam, this is normalized projected raw-gradient change along the realized path; it is not the SGD directional-curvature estimate. A value above two requires negative cosine and a sufficiently large consecutive norm ratio. Thus, reversal and norm growth explain the diagnostic's increase algebraically, rather than supplying independent confirmations of instability. The reference at two recalls the GD analysis, but does not identify an Adam stability boundary.

A. Controlled Full-Batch Training

We reuse the CIFAR-10/MLP, CIFAR-10/ResNet-32 and WikiText-2/Transformer objectives from Section IV. Each Adam trajectory starts from the same frozen, untrained seed-0 weights used for its GD comparison, with fresh moments. We run 1,024 full-objective updates at each constant rate: $\{0.0001, 0.0003, 0.001\}$ for the MLP and

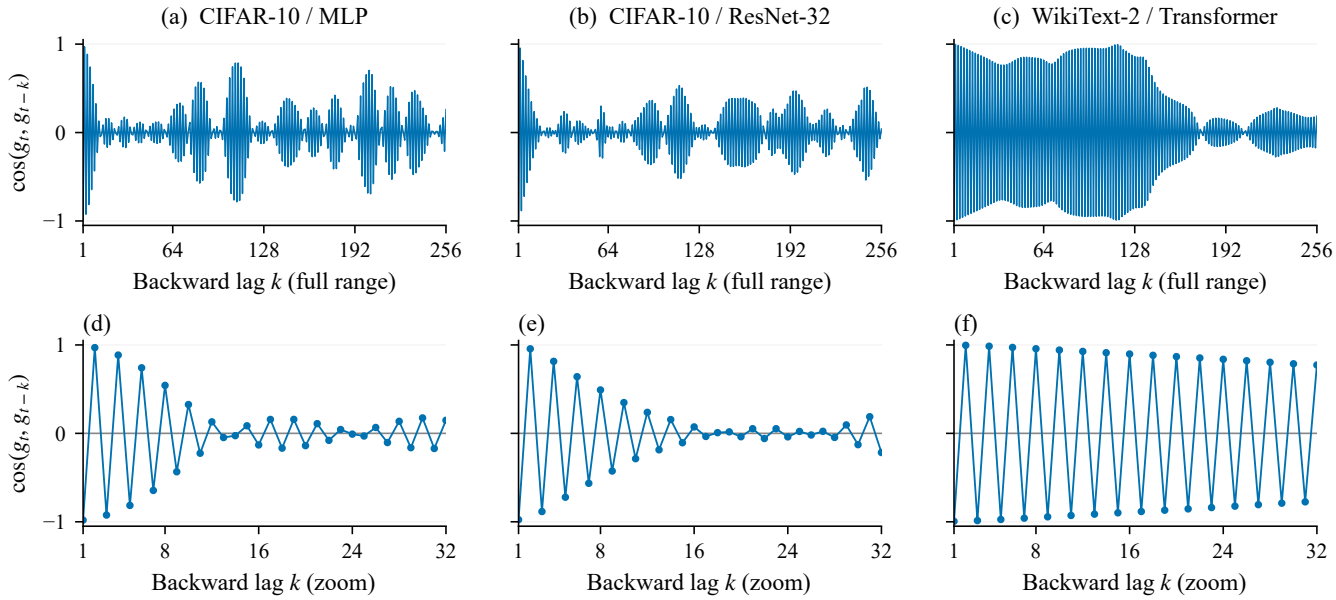


Fig. 3: Backward gradient alignment at the common endpoint state θ_{2256}, corresponding to $t = 2257$ under the manuscript's preupdate gradient indexing. Each point is one exact cosine $\cos(g_t, g_{t-k})$; the endpoint is fixed across all 256 lags. The top row shows lags 1–256; the bottom row magnifies the same values at lags 1–32 to resolve the early sign alternation. Rates are 0.04 (MLP), 0.1 (ResNet-32) and 0.2 (Transformer). These are separately recomputed 256-update GD continuations from retained state 2000, with one initialization and no smoothing or endpoint averaging. Appendix B records the additional endpoints and numerical replay scope.

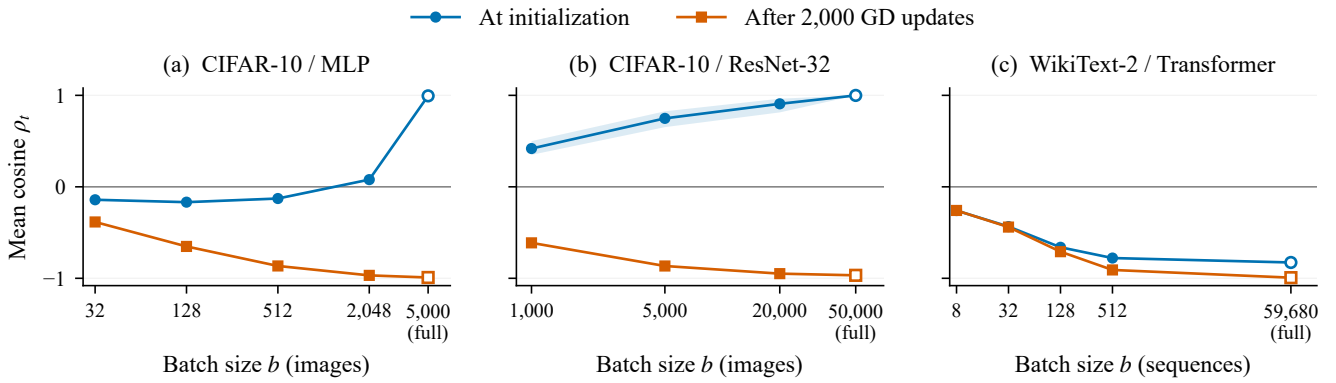


Fig. 4: Mean sampled-gradient cosine over 199 adjacent pairs during 200-update continuations in three model–dataset cases. Blue circles start at initialization; orange squares start after 2,000 GD updates. Each line averages repetitions within initialization, then weights the three initialization blocks equally; shading is their range, not a confidence interval. Hollow rightmost markers denote full GD. ResNet batches contain whole fixed 1,000-image normalization groups; Transformer batches count prepared sequences. Batch axes are logarithmic; lines guide the eye between measured sizes.

$\{0.0003, 0.001, 0.003\}$ for ResNet and Transformer. All nine arms use $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$, without a schedule, clipping or weight decay. There is one initialization per model, matched across rates. Full-batch accumulation removes sampling variation in the objective; model, objective and numerical details are in Appendix B-C.

Figure 5 shows overall convergence interspersed with transient loss spikes, especially at larger rates. The largest tested rate attains the lowest final training loss in each model despite these excursions. A spike can therefore interrupt a trajectory that continues to improve; its presence alone does not rank the final loss. We next examine the gradient dynamics within individual excursions before summarizing their distribution across the nine arms.

B. Anatomy of Individual Spikes

For each model, we select the recovered excursion with the largest peak/pre-rise loss ratio at its highest tested rate. Selection uses only loss, with no condition on a diagnostic. The selected peaks occur at states 703, 422 and 794 for MLP,

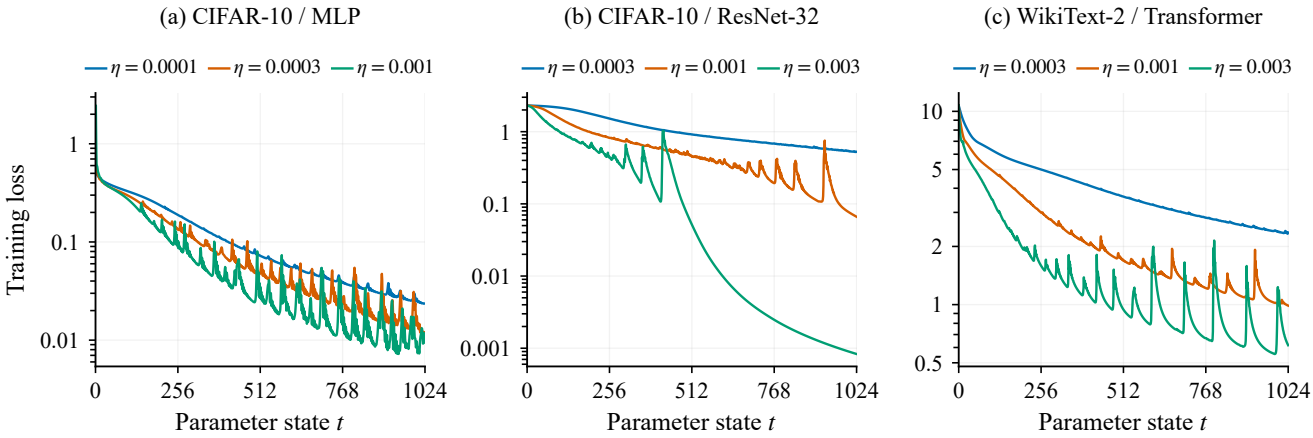


Fig. 5: Full-batch Adam training from matched initial weights, with three constant learning rates per model. Every training-loss observation is shown without smoothing on logarithmic y-axes. Parameter state s denotes s completed updates, from initialization to 1,024 updates. One initialization per model; colors identify increasing rates within each panel.

ResNet and Transformer, respectively. Their location in the middle or later part of training is an outcome of this maximum-amplitude rule, not an additional temporal filter.

All three examples in Fig. 6 enter the pronounced rise with negative raw-gradient cosine. During the rise, the gradient norm increases sharply and $\eta\widehat{\kappa}$ exceeds two before the loss maximum. Momentum norms also grow, with model-dependent timing relative to the raw gradient. In the Transformer, the first crossing of two within the excursion is at state 788, when loss is only 0.40% above its pre-rise level. The +10% crossing follows at 790; raw-gradient and momentum norms reach their event maxima at 792 and 793, before loss peaks at 794. The diagnostic thus precedes the pronounced spike and its maximum, while the first small loss increase has already occurred.

The descent separates the models. ResNet and Transformer move toward positive gradient and momentum cosines, lower projected change and a comparatively smooth loss decline. The MLP continues to oscillate: raw cosine remains mostly negative while momentum cosine changes sign, and loss and norms decrease through repeated rebounds. Recovery therefore need not require every raw-gradient pair to become constructive.

These examples associate the developing loss spike with gradient reversal and rapid norm amplification, even though the parameter steps are adaptive. An above-two value marks a strong local gradient response consistent with the observed excursion; it is not proof of a linear instability of Adam. The early response suggests an opportunity to anticipate further growth and intervene. Whether it supplies a reliable trigger, or whether acting on it improves training, requires evidence beyond selected examples.

C. Loss-Defined Events and Their Distribution

To examine repetition without selecting on gradients, write $\mathcal{L}_s := \mathcal{L}(\theta_s)$ for the recorded training loss. Starting at initialization, let a_i be the earliest available state with $\mathcal{L}_{a_i+1} > \mathcal{L}_{a_i}$. Hold the pre-rise reference $\mathcal{L}_{a_i}$ fixed. When a return is observed within T updates, define the first recovery and peak by

$$\begin{aligned} c_i &:= \min\{s : a_i < s \le T,\ \mathcal{L}_s \le \mathcal{L}_{a_i}\}, \\ p_i &:= \min \arg\max_{a_i<s<c_i} \mathcal{L}_s. \end{aligned} \tag{28}$$

An event is included only when recovery is observed by $T = 1024$. Search resumes at c_i; partial declines and additional local maxima before this return belong to the same event. Thus a_i is the last pre-rise state, $a_i + 1$ the first elevated state, p_i the loss peak, and c_i the first recovered state. Peak increase is $100(\mathcal{L}_{p_i}/\mathcal{L}_{a_i} - 1)$; rise and recovery times are $p_i - a_i$ and $c_i - p_i$, in optimizer updates. The three loss ratios are 1, $\mathcal{L}_{p_i}/\mathcal{L}_{a_i}$ and $\mathcal{L}_{c_i}/\mathcal{L}_{a_i} \le 1$.

Figure 7 places the illustrations within the complete distribution of recovered excursions. Small fluctuations coexist with long tails; larger rates populate more severe excursions, although the distributions overlap and do not undergo a uniform shift. To focus the anatomy analysis on pronounced changes, we retain events with peak increase strictly greater than 10%. This is an operational amplitude filter chosen for interpretation, not a separation inferred from a density minimum or an optimizer stability threshold. The density retains all amplitudes so that the selection remains visible in context. The filtered sample contains 129 recovered events. Two further above-threshold candidates without observed recovery are excluded.

For Table II, pre-rise diagnostics are measured at θ_{a_i}, hence use gradient index $a_i + 1$. Recovery entries first average each event's diagnostic over the arrival states $p_i + 1, \ldots, c_i$, then summarize those means across events with equal event weight. They are not values only at c_i. The norm-ratio columns instead use the endpoints $\|g_{p_i+1}\|/\|g_{a_i+1}\|$ and $\|g_{c_i+1}\|/\|g_{p_i+1}\|$. The gradient norm's own maximum need not coincide with the loss peak.

Among qualifying events, both the mean increase and the maximum increase grow with LR in every model with more than one eligible rate. The largest increase is 858.8% for ResNet at $\eta = 0.003$. Mean recovery takes 17–19 updates

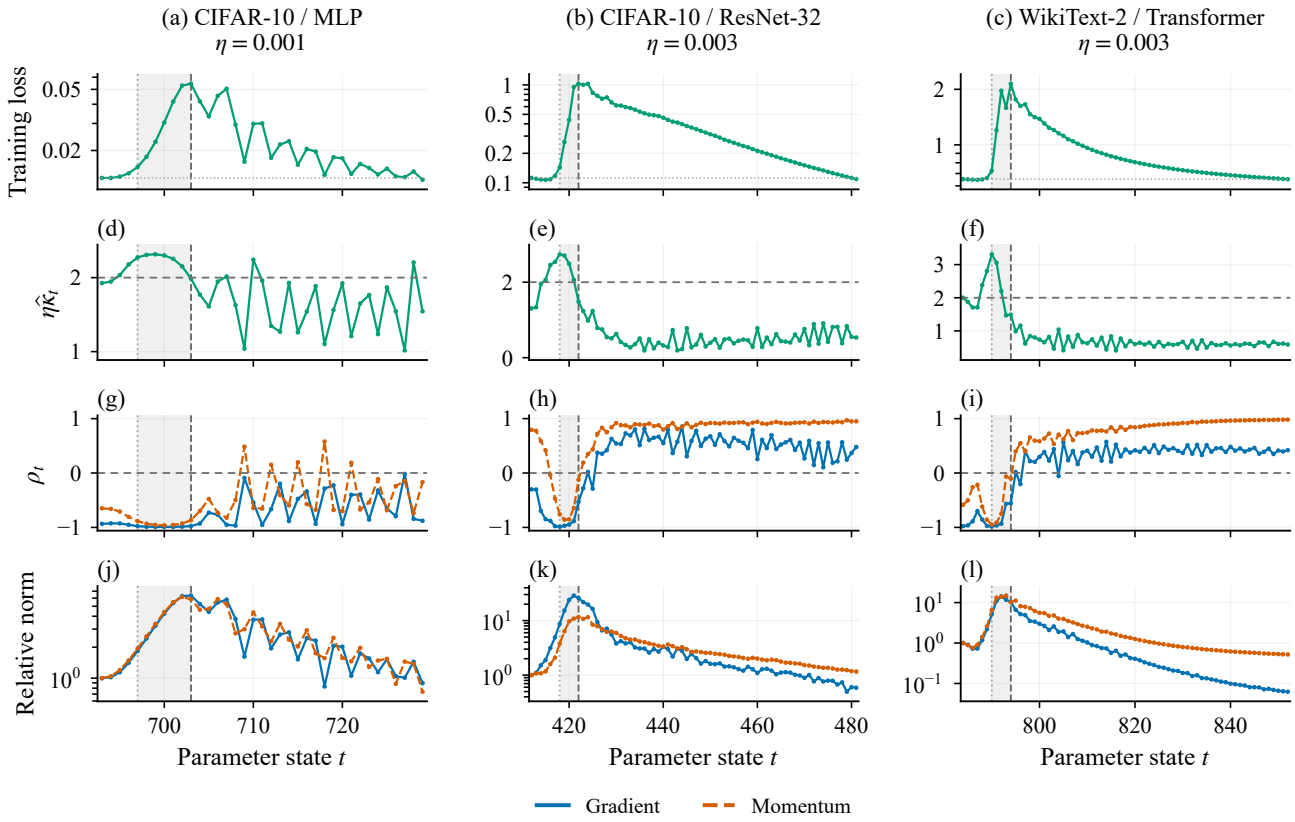


Fig. 6: Largest recovered relative loss spike at each model's highest Adam rate. Rows show loss, raw-gradient projected change, consecutive cosines of raw gradients and first moments, and their relative norms. All observations are unsmoothed. At parameter state s, vector diagnostics refer to g_{s+1} and m_{s+1}. Norms are divided by their value at the first displayed state. Gray bands mark the pronounced rise from the first +10% crossing to the loss peak (697–703, 418–422 and 790–794), rather than the entire excursion through recovery. The horizontal dotted loss reference is the first displayed loss. Windows begin ten states before the peak and end at the first later loss below that displayed reference.

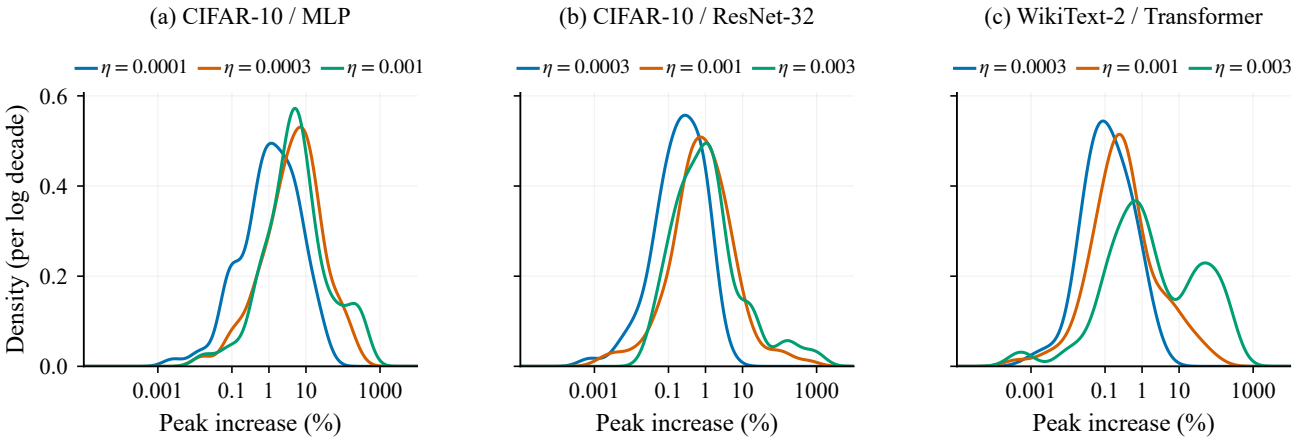


Fig. 7: Distributions of peak percentage increases for all 806 recovered excursions, including small fluctuations, with no amplitude cutoff. Colors identify LRs within each model. Gaussian kernel densities are estimated in $\log_{10}$ percentage-increase coordinates with one pooled bandwidth per model, common across its LRs. Each LR density integrates to one in that coordinate. The log x-axis shows percentage increase: 1 denotes +1%, and 100 denotes +100%. This differs from the loss-ratio axis in Fig. 8.

TABLE II: Anatomy of recovered spikes with peak increase greater than 10%. Cells report mean ± sample SD across events; maximum peak increase has its own subcolumn. Before denotes a single diagnostic at state a_i; recovery denotes the event's mean over states $p_i+1,\dots,c_i$. Norm ratios compare loss-peak/pre-rise and recovered/loss-peak endpoints. N counts included events; dashes indicate no qualifying event. Two initialization events have undefined pre-rise diagnostics, leaving 46 and 38 observations for those entries in the MLP's middle and highest LR rows. No smoothing or averaging across initializations.

Dataset Model	η	N	Peak increase (%)		Time (updates)		$\eta\widehat{\kappa}$ (raw)		ρ (raw)		Gradient norm ratio	
			Mean ± SD	Max	Rise	Recovery	Before	Recovery	Before	Recovery	Rise	Recovery
CIFAR-10 MLP	10^{-4}	8	19 ± 6.1	31	8.4 ± 2.9	4.4 ± 2.4	1.8 ± 0.38	1.7 ± 0.08	-0.77 ± 0.31	-0.89 ± 0.08	5.3 ± 1.7	0.30 ± 0.07
	3×10^{-4}	47	43 ± 46	203	5.5 ± 4.4	4.9 ± 4.0	1.3 ± 0.25	1.6 ± 0.18	-0.53 ± 0.29	-0.71 ± 0.24	4.5 ± 3.0	0.33 ± 0.13
	10^{-3}	39	95 ± 107	318	6.1 ± 5.9	7.4 ± 7.5	1.4 ± 0.40	1.4 ± 0.26	-0.46 ± 0.38	-0.55 ± 0.30	5.2 ± 3.8	0.31 ± 0.14
CIFAR-10 ResNet-32	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	10	105 ± 182	602	7.5 ± 2.2	17 ± 11	1.3 ± 0.39	1.3 ± 0.25	-0.36 ± 0.42	-0.36 ± 0.26	4.3 ± 2.9	0.43 ± 0.24
	3×10^{-3}	7	179 ± 309	859	6.3 ± 1.3	19 ± 20	1.7 ± 0.38	1.1 ± 0.37	-0.65 ± 0.28	-0.13 ± 0.42	5.0 ± 4.2	0.34 ± 0.24
WikiText-2 Transformer	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	6	33 ± 25	81	6.3 ± 1.9	24 ± 9.8	2.1 ± 0.27	0.96 ± 0.16	-0.84 ± 0.10	-0.02 ± 0.12	9.0 ± 3.8	0.09 ± 0.05
	3×10^{-3}	12	82 ± 70	231	6.2 ± 2.7	31 ± 17	2.2 ± 0.43	0.91 ± 0.19	-0.79 ± 0.22	0.09 ± 0.21	11 ± 6.9	0.08 ± 0.08

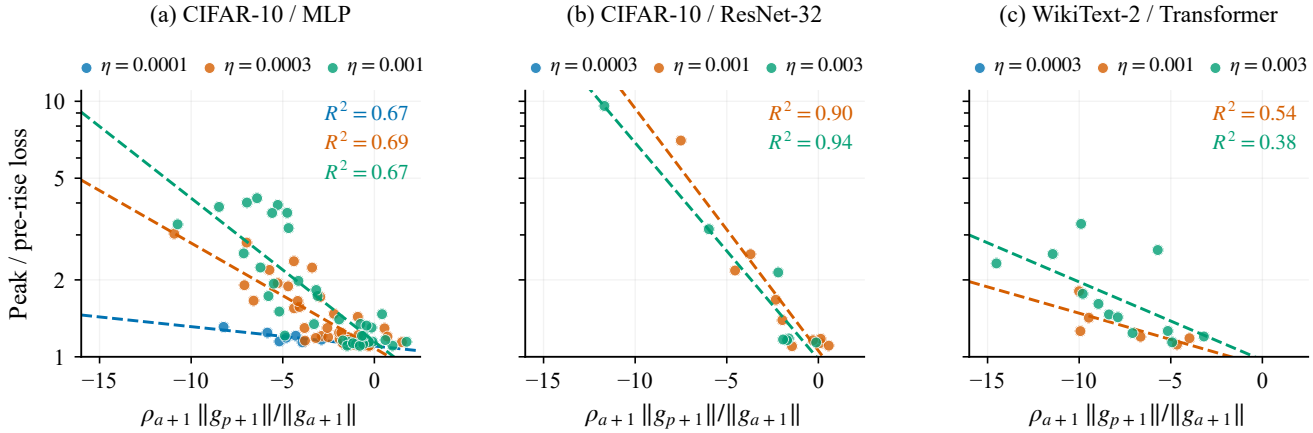


Fig. 8: Pre-rise raw-gradient cosine times rise norm ratio versus peak/pre-rise loss for 127 recovered spikes above 10%; two initialization events have undefined pre-rise cosine. One unsmoothed point per event. Colors identify LRs. Separate per-LR least-squares fits with intercept use $\log_{10}$(peak/pre-rise loss) as the response; color-matched R^2 values are descriptive in-sample scores on that log response. Dashed lines extend across each panel, including extrapolation beyond observed samples. The y-axis is a loss ratio: 1 denotes no increase and 2 denotes +100%.

versus 6–8 for the ResNet rise, and 24–31 versus about 6 for the Transformer. MLP rise and recovery times are more comparable. Thus, a short amplification can be followed by a much longer descent, without implying that all intermediate loss changes are monotonic.

Pre-rise raw cosine is negative in 117 of 127 defined events, and the gradient norm at the loss peak exceeds its pre-rise value in all 129. Projected change is already above two before the first loss increase in 21 of 127 defined cases. Its full-event maximum precedes the loss peak in 82 of 129 events, coincides in 29 and follows in 18. These observations support recurrent sign-and-magnitude structure, rather than a universal pre-onset crossing. Recovery is also heterogeneous: event-averaged raw cosines remain negative for the MLP and the eligible ResNet groups, and are near zero for the Transformer. The positive recovery segments in the illustrations therefore do not describe every event.

D. Reversal, Norm Amplification and Possible Interventions

To relate the initial regime to amplification during an event, we plot

$$\rho_{a_i+1}\frac{\|g_{p_i+1}\|}{\|g_{a_i+1}\|} \quad \text{against} \quad \frac{\mathcal{L}_{p_i}}{\mathcal{L}_{a_i}}. \tag{29}$$

This combines the pre-rise consecutive cosine with the subsequent rise norm ratio. It is an event descriptor, not Eq. (27), which pairs a cosine with a *consecutive* norm ratio. Its magnitude incorporates the future peak and cannot be used unchanged as an online predictor.

Figure 8 relates larger spikes to the combination of a destructive initial gradient regime and norm amplification. The per-rate fits explain 67–69% of log-loss-ratio variation for the MLP, 90–94% for ResNet, and 38–54% for the Transformer. The association is consistent with the qualitative anatomy, with substantial remaining model-dependent dispersion. It does not

establish a linear relation on the original loss scale, an Adam stability criterion or a causal feedback loop. Both axes use loss-defined endpoints, the sample is conditional on future amplitude and recovery, and events within a trajectory can be dependent.

This anatomy motivates testing whether intervention during reversal and early norm growth can attenuate the subsequent excursion. Gradient-norm clipping was proposed to address exploding gradients [29], and analyses allowing smoothness to grow with gradient norm provide a further rationale for adaptive scaling [30]. Our observations are consistent with investigating norm control, without identifying those theoretical conditions or establishing that clipping would help these Adam runs. Clipping Adam's input changes both moment streams; its effect on the adaptive step cannot be inferred from the raw norm alone. Learning-rate adjustment and data-mixing interventions provide other tests of the proposed interaction between the current regime and subsequent gradient growth. These tests should isolate states before pronounced excursions, compare matched interventions with unchanged continuations, and measure both spike attenuation and later loss at equal budgets. Establishing prospective warning information also requires non-spike comparisons. Such interventions and predictive validation remain future work.

VI. Conclusion

The displacement formulation distinguishes the magnitude of individual updates, their local directional interaction and their cumulative accumulation. For plain SGD, the exact decomposition retains all temporal lags, while consecutive cosine provides a tractable local observation. The curvature and sampling relations clarify how that observation can change without determining either cumulative displacement or loss improvement. The full-batch and mini-batch experiments illustrate these distinctions: gradient reversal can coexist with objective progress, and weaker reversal need not improve displacement retention or the attained loss.

Under Adam, raw-gradient diagnostics remain informative about the loss landscape traversed by adaptive steps. Controlled full-batch experiments connect loss spikes with a predominantly destructive pre-rise gradient regime and sharp norm amplification. The micro-plots resolve projected gradient changes before the loss maximum; the event population quantifies the variation in timing and recovery across models and rates. A joint descriptor of pre-rise reversal and rise norm growth is associated with relative peak magnitude, without establishing a causal feedback mechanism or a universal instability threshold.

The practical contribution is a framework for interpreting these measurements together and using them to formulate controlled interventions. Developing a scheduler or clipping rule requires demonstrating that acting on early diagnostics improves subsequent loss at matched budgets, beyond decisions based on recent loss and known training settings. Prospective validation, controlled interventions and replication across independent runs remain future work.

Appendix A
Curvature Estimation and Local Loss Change

Write $H(\theta) = \nabla^2 \mathcal{L}(\theta)$ and assume the objective is twice continuously differentiable along the plain-SGD step. Integrate its gradient derivative on $\theta_{t-2} - s\eta g_{t-1}$, add the endpoint-noise difference, and project onto $-g_{t-1}/(\eta\|g_{t-1}\|^2)$. The result is

$$\widehat{\kappa}_t = \int_0^1 \frac{g_{t-1}^\top H(\theta_{t-2} - s\eta g_{t-1}) g_{t-1}}{\|g_{t-1}\|^2}\, ds - \frac{\varepsilon_t}{\eta}. \tag{30}$$

If H is L_H-Lipschitz in operator norm on the step, its full-batch specialization satisfies

$$|\widehat{\kappa}_t - \kappa_t| \le \int_0^1 L_H s\eta \|g_{t-1}\|\, ds = \frac{L_H \eta}{2}\|g_{t-1}\|. \tag{31}$$

This bounds the difference between the measured path average and the starting-point directional curvature. For Adam, integrate along its adaptive step and project onto the preceding raw gradient. This gives the exact path-average form of Eq. (27).

Consider $\theta_{t-1} = \theta_{t-2} - \eta g_{t-1}$ and write $\delta\mathcal{L}_{t-1} = \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. Integrating the full gradient along the step gives

$$\delta\mathcal{L}_{t-1} = -\eta \int_0^1 g_{t-1}^\top \nabla\mathcal{L}(\theta_{t-2} - s\eta g_{t-1})\, ds. \tag{32}$$

The trapezoidal approximation replaces the integrand by its endpoint average. Substituting $\nabla\mathcal{L}(\theta_{t-2}) = g_{t-1} - \xi_{t-1}$ and $\nabla\mathcal{L}(\theta_{t-1}) = g_t - \xi_t$ yields

$$\begin{aligned}\delta\mathcal{L}_{t-1} \approx -\frac{\eta}{2}\left(\|g_{t-1}\|^2 + g_{t-1}^\top g_t\right) \\ + \frac{\eta}{2} g_{t-1}^\top(\xi_{t-1} + \xi_t).\end{aligned} \tag{33}$$

The substitution $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ recovers Eq. (22). The expression is exact on quadratic objectives. A locally Lipschitz Hessian bounds its error by a constant times $\|\eta g_{t-1}\|^3$.

Using $\rho_t\|g_t\|/\|g_{t-1}\| \approx 1 - \eta\kappa_t + \varepsilon_t$ and Eq. (14) gives the equivalent curvature form,

$$\begin{aligned}\delta\mathcal{L}_{t-1} \approx -\eta\|g_{t-1}\|^2\left(1 - \frac{\eta\kappa_t}{2}\right) \\ + \eta\xi_{t-1}^\top g_{t-1}.\end{aligned} \tag{34}$$

It also follows directly from a second-order Taylor expansion at θ_{t-2}. With zero noise, the conditions for a loss-increasing step reduce to Eqs. (23) and (24).

The noise term has a simple conditional interpretation. Given θ_{t-2}, unbiased sampling gives $\mathbb{E}[\|g_{t-1}\|^2] = \|\nabla\mathcal{L}(\theta_{t-2})\|^2 + \mathbb{E}[\|\xi_{t-1}\|^2]$ and $\mathbb{E}[\xi_{t-1}^\top g_{t-1}] = \mathbb{E}[\|\xi_{t-1}\|^2]$. Their first-order variance terms cancel, leaving

$$\begin{aligned}\mathbb{E}[\delta\mathcal{L}_{t-1} \mid \theta_{t-2}] \approx -\eta\|\nabla\mathcal{L}(\theta_{t-2})\|^2 \\ + \frac{\eta^2}{2}\mathbb{E}[g_{t-1}^\top H_{t-1} g_{t-1} \mid \theta_{t-2}].\end{aligned} \tag{35}$$

Thus sampling contributes through curvature at second order. A nonlinear objective also contributes the finite-step remainder.

APPENDIX B
ESSENTIAL MEASUREMENT CONVENTIONS

A. Fixed Objectives and Numerical Scope

The tanh MLP has two 200-unit hidden layers and minimizes half-squared error summed over ten outputs and averaged over the first 5,000 CIFAR-10 images. ResNet-32 minimizes mean cross-entropy over all 50,000 training images, using 50 fixed groups of 1,000 for batch normalization. Standardization uses the complete CIFAR-10 training split: coordinate-wise for the MLP and channel-wise for ResNet. Full-batch accumulation evaluates the same fixed objective at every update; augmentation and dropout are disabled.

The word-level Transformer has two layers, two heads and embedding/feed-forward dimensions of 200. Twenty corpus streams form causal chunks of at most 35 tokens, yielding 59,680 sequences and 2,088,600 supervised targets. The objective averages negative log-likelihood over valid targets, excluding padding. Architecture and the 33,278-word first-occurrence dictionary over training, validation and test text follow PyTorch example revision `13acec6d7c78` [31].

Diagnostics cover all trainable parameters. Parameters, gradients and Adam moments use FP32; reductions use FP64, with TF32 disabled. Initial unpaired cosines and zero-denominator ratios are undefined. The GD plots' trailing means summarize sustained regimes and can delay a displayed transition; Adam losses and micro-plots are unsmoothed. All claims concern training objectives. Matched tensors give one initialization per full-batch model; independent hardware executions can develop finite-precision drift.

B. Sampling and Trajectory Comparisons

Stochastic batches are independent uniform subsets without replacement; within a repetition, batch sizes share prefixes of a common permutation. ResNet samples complete normalization groups. Transformer sampling uses $N = 59{,}680$ sequences and $M = 2{,}088{,}600$ total targets: for sequence summed losses ℓ_i, the estimator is $N(bM)^{-1}\sum_{i\in B}\ell_i$. Its fixed denominator preserves the full token-weighted objective. The seed-2 smallest two ResNet batches use a different runtime, with 200-update full-GD replay parameter relative error 2.36×10^{-7}; batch and runtime remain confounded in this block. The backward-lag curves also use separate continuations, whose later floating-point paths can differ from the original GD trajectories.

C. Adam Event Summaries

At state θ_s, the raw gradient and updated first moment are g_{s+1} and m_{s+1}, before parameter update $s+1$. The terminal raw gradient is evaluated without another update. Hardware varies across ResNet and Transformer learning-rate arms; initial-gradient agreement is checked, while subsequent finite-precision drift remains possible. Rate selection uses training calibration.

Two excursions above 10% without observed recovery are excluded. Two retained initialization events have undefined pre-rise diagnostics. Each event has equal weight; recovery diagnostics first average within the event, and sample SD uses denominator $n-1$. The scatter uses per-rate ordinary least squares with an intercept on log relative peak loss. Its R^2 describes the fitting sample; dashed extensions extrapolate. For the density, each model's pooled log-percentage SD times pooled $n^{-1/5}$ sets the common Gaussian bandwidth. Events within a trajectory can be dependent.

REFERENCES

[1] J. Hestness, S. Narang, N. Ardalani, G. Diamos, H. Jun, H. Kianinejad, M. Patwary, Y. Yang, and Y. Zhou, "Deep learning scaling is predictable, empirically," *arXiv preprint arXiv:1712.00409*, 2017.

[2] J. Kaplan, S. McCandlish, T. Henighan *et al.*, "Scaling laws for neural language models," *arXiv preprint arXiv:2001.08361*, 2020.

[3] T. Henighan *et al.*, "Scaling laws for autoregressive generative modeling," *arXiv preprint arXiv:2010.14701*, 2020.

[4] J. Hoffmann, S. Borgeaud, A. Mensch, E. Cassirer, T. Jones, T. Dasilva, K. Tunyasuvunakool, C. Humphreys, E. Rutherford, E. Buchatskaya *et al.*, "Training compute-optimal large language models," *arXiv preprint arXiv:2203.15556*, 2022.

[5] P. Goyal, P. Dollár, R. Girshick, P. Noordhuis, L. Wesolowski, A. Kyrola, A. Tulloch, Y. Jia, and K. He, "Accurate, large minibatch sgd: Training imagenet in 1 hour," *arXiv preprint arXiv:1706.02677*, 2017.

[6] L. N. Smith, "Cyclical learning rates for training neural networks," in *2017 IEEE winter conference on applications of computer vision (WACV)*. IEEE, 2017, pp. 464–472.

[7] I. Loshchilov and F. Hutter, "Sgdr: Stochastic gradient descent with warm restarts," *arXiv preprint arXiv:1608.03983*, 2016.

[8] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *International conference on machine learning*. pmlr, 2015, pp. 448–456.

[9] A. Krogh and J. Hertz, "A simple weight decay can improve generalization," *Advances in neural information processing systems*, vol. 4, 1991.

[10] B. T. Polyak, "Some methods of speeding up the convergence of iteration methods," *USSR Computational Mathematics and Mathematical Physics*, vol. 4, pp. 1–17, 1964.

[11] D. P. Bertsekas, *Nonlinear Programming*, 2nd ed. Belmont, MA: Athena Scientific, 1999.

[12] S. Jastrzębski, M. Szymczak, S. Fort, D. Arpit, J. Tabor, K. Cho, and K. Geras, "The break-even point on optimization trajectories of deep neural networks," in *International Conference on Learning Representations (ICLR)*, 2020.

[13] J. M. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar, "Gradient descent on neural networks typically occurs at the edge of stability," in *International Conference on Learning Representations (ICLR)*, 2021.

[14] S. Arora, Z. Li, and A. Panigrahi, "Understanding gradient descent on the edge of stability in deep learning," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 2022, pp. 948–1024.

[15] A. Damian, E. Nichani, and J. D. Lee, "Self-stabilization: The implicit bias of gradient descent at the edge of stability," in *International Conference on Learning Representations (ICLR)*, 2023.

[16] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[17] P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur, "Sharpness-aware minimization for efficiently improving generalization," in *International Conference on Learning Representations (ICLR)*, 2021.

[18] L. Sagun, U. Evci, V. U. Guney, Y. Dauphin, and L. Bottou, "Empirical analysis of the hessian of over-parametrized neural networks," 2017.

[19] B. Ghorbani, S. Krishnan, and Y. Xiao, "An investigation into neural net optimization via hessian eigenvalue density," in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 2232–2241.

[20] S. Mandt, M. D. Hoffman, and D. M. Blei, "Stochastic gradient descent as approximate bayesian inference," *Journal of Machine Learning Research*, vol. 18, no. 134, pp. 1–35, 2017. [Online]. Available: https://www.jmlr.org/papers/v18/17-214.html

[21] W. Hu, C. J. Li, L. Li, and J.-G. Liu, "On the diffusion approximation of nonconvex stochastic gradient descent," *Annals of Mathematical Sciences and Applications*, vol. 4, no. 1, pp. 3–32, 2019.

[22] S. Mandt, M. D. Hoffman, and D. M. Blei, "Stochastic gradient descent as approximate bayesian inference," *Journal of Machine Learning Research*, vol. 18, no. 1, pp. 4873–4907, 2017.

[23] S. Jastrzębski, Z. Kenton, D. Arpit, N. Ballas, A. Fischer, Y. Bengio, and A. Storkey, "Three factors influencing minima in sgd," *arXiv preprint arXiv:1711.04623*, 2017. [Online]. Available: https://arxiv.org/abs/1711.04623

[24] A. Chowdhery, S. Narang, J. Devlin, M. Bosma, G. Mishra, A. Roberts *et al.*, "PaLM: Scaling language modeling with pathways," *arXiv preprint arXiv:2204.02311*, 2022. [Online]. Available: https://arxiv.org/abs/2204.02311v5

[25] S. Zhang, S. Roller, N. Goyal, M. Artetxe, M. Chen, S. Chen *et al.*, "OPT: Open pre-trained transformer language models," *arXiv preprint arXiv:2205.01068*, 2022. [Online]. Available: https://arxiv.org/abs/2205.01068v4

[26] Team OLMo, P. Walsh, L. Soldaini, D. Groeneveld, K. Lo *et al.*, "2 OLMo 2 furious," *arXiv preprint arXiv:2501.00656*, 2025. [Online]. Available: https://arxiv.org/abs/2501.00656v3

[27] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle *et al.*, "The Llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024. [Online]. Available: https://arxiv.org/abs/2407.21783v3

[28] J. Z. HaoChen, C. Wei, J. Lee, and T. Ma, "Shape matters: Understanding the implicit bias of the noise covariance," in *Conference on Learning Theory*. PMLR, 2021, pp. 2315–2357.

[29] R. Pascanu, T. Mikolov, and Y. Bengio, "On the difficulty of training recurrent neural networks," in *International conference on machine learning*. Pmlr, 2013, pp. 1310–1318.

[30] J. Zhang, T. He, S. Sra, and A. Jadbabaie, "Why gradient clipping accelerates training: A theoretical justification for adaptivity," in *International Conference on Learning Representations*, 2020. [Online]. Available: https://openreview.net/forum?id=BJgnXpVYwS

[31] PyTorch contributors, "PyTorch examples: Word language model," GitHub repository, revision `13acec6d7c78`; full commit specified in the URL. [Online]. Available: https://github.com/pytorch/examples/tree/13acec6d7c78dacd5e1fe9b0b4a325e1d39abc15/word_language_model